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IDEAS TO IMPACT



GREEN HYDROGEN PASSENGER MOBILITY

Trilogic 2.0
Motilal Nehru National Institute of Technology Allahabad, Prayagraj

05.OCTOBER.2026

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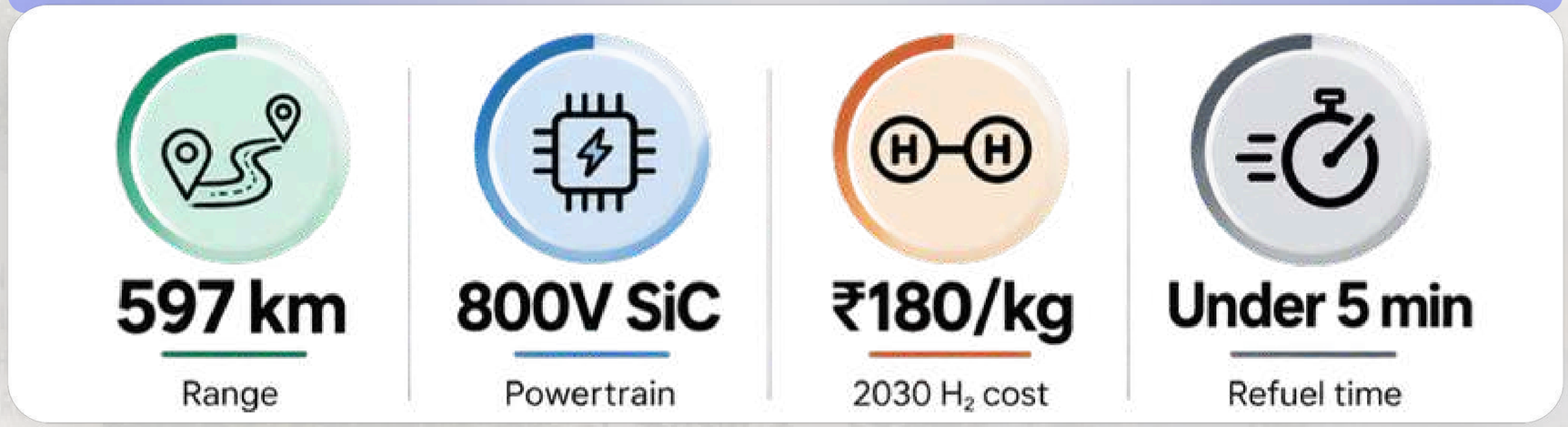
"I travel in a hydrogen car. It is better than Mercedes."

- Nitin Gadkari, Union Minister of Road Transport and Highways, July 2026

FCEVs will not replace BEVs; they will complement them.

A complete ₹19,744 Cr NGHM-backed supply chain roadmap

from production to refuelling to a vehicle engineered around India's own hydrogen mission, not a generic global platform



- **Not a lab concept-** Benchmarked against a vehicle under a two-year NISE trial(Toyota Mirai, NISE 2025 trial)
- **Not hoped-for infrastructure-** Built on hydrogen corridors India launched in July 2026
- **Not a policy assumption-** Priced on the same 5% GST rate BEVs already get

"The Country already believes it. We proposed the vehicle, the supply chain, and the economics to prove it right."

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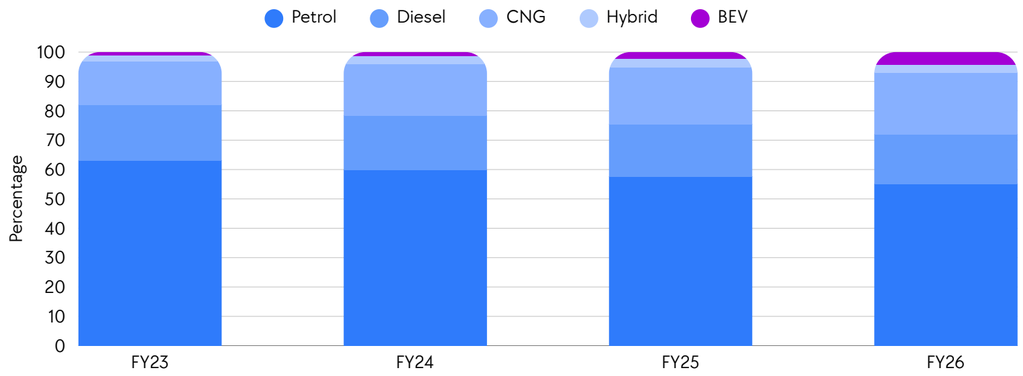
India's Mobility Challenge: Is One Pathway Enough?

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Market Reality (The Powertrain Mix & BEV Growth charts)

India PV Market — Powertrain Mix

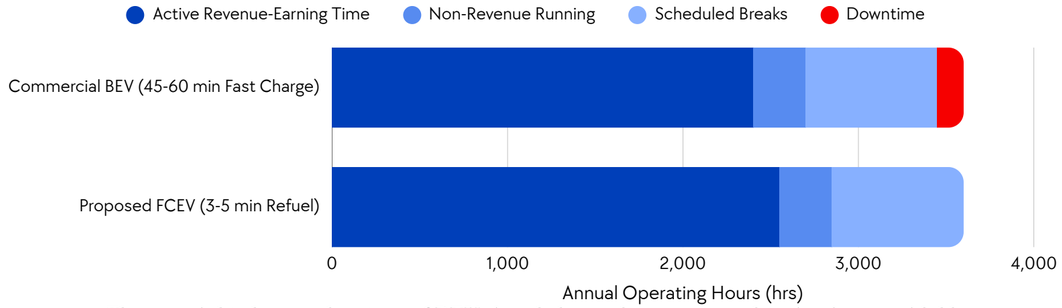
Conventional powertrains still account for more than 90% of India's 4.6M+ unit market.



Source — Vahan Dashboard / EVreporter / JMK Research.

The market is huge but dominated by Conventional Drivetrains

nnual Fleet Productivity Impact: BEV vs. FCEV



The overnight charge advantage of BEVs is ruled out when en route charging is unavoidable, as is the case with the high utilization intercity fleets with multi-day assignments.

Daily fleet revenue = Duty hours x Utilization(%) x Fare Rate

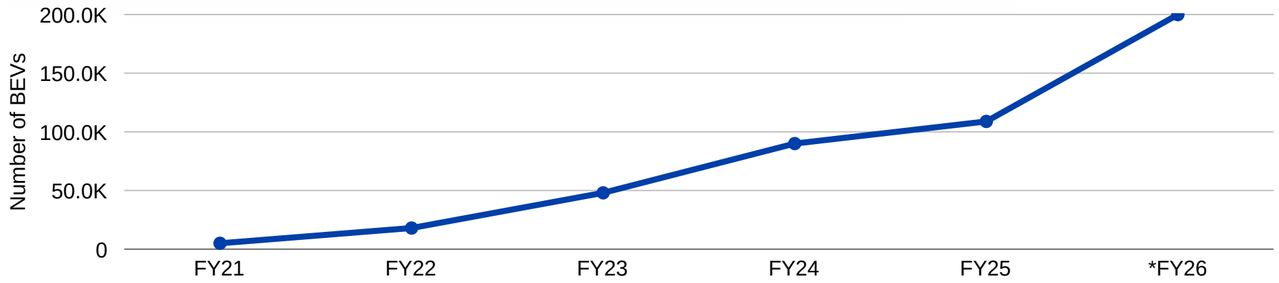
↓ Operating Hours ↓ Fleet Productivity

The Indian Hydrogen Mandate

- **National Green Hydrogen Mission (NGHM)**: ₹19,744 Cr government outlay approved.
- **Production Target**: 5 MMT of green H_2 capacity per annum by 2030.
- **Mobility Focus**: Explicit financial allocation by MNRE for surface transport pilot projects.

BEV Adoption on The Rise

BEV growth proves the Indian consumer's appetite for zero-emission mobility.



Source: Vahan Dashboard via EVreporter / JMK Research.

BEV adoption is already rising. Hydrogen is not proposed as a replacement for electrification but as an additional zero-emission powertrain option where BEV has practical limitations.

THE HYDROGEN CASE: WHY IT MATTERS FOR INDIA?

The Gravimetric Advantage (Range without the Weight): FCEVs offer competitive, long-range mobility (597 km) without the massive curb-weight penalty of extremely large battery packs, preserving vehicle dynamics and efficiency.

High-Utilization & Low Downtime: For intercity travel or commercial fleet use-cases (like taxis/Tour segments), FCEVs eliminate 30 - 60 minute charging bottlenecks. A ~5-minute refuel maximizes asset profitability.

Shared Refuelling Infrastructure: Hydrogen refuelling infrastructure is shared between FCEVs and light-mobility H2-ICE vehicles across the industry, including Suzuki's own Swift H2-ICE and Burgman H2 prototypes, amortising infrastructure cost across vehicle classes.

Target Segment

India's intercity taxi + high-utilization commercial fleet segment (300-350 km/day) the exact segment where a 60-min charging stop costs a fleet operator real money every single day.

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PROPULSION SELECTION: The Physics of the D-Segment

H_2-ICE suits lightweight retrofit applications; for our 1,950 kg GVW long-range sedan, PEMFC offers lower H_2 consumption with zero combustion NO_x at the tailpipe.

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PROPULSION TRADE-OFF MATRIX

Criteria	Battery Electric (BEV)	H2 Internal Combustion (H2-ICE)	Proposed PEM FCEV
Tank-to-Wheel Efficiency	**80–85%** (Best urban)	32–38%	**48.1%** (Ideal for long-range)
System Curb Weight Penalty	High (+400 kg battery pack)	Medium (Engine + 700-bar tank)	**Low (1,450 kg target mass)**
Tailpipe Emissions	Zero	Trace of NOx	**Zero (pure H_2O vapor)**
Refueling / Charge Speed	45–60 min (Fast charge)	3–5 min	**3–5 min (SAE J2601)**
Primary Use-Case	Intra-city / Short Commutes	Retrofit	**Intercity Fleet / Taxi (300-350 km/day)**

THE THERMODYNAMIC CORE: WHY FCEV IS SUPERIOR

H2-ICE: THERMAL LIMITATION

Bound by Otto Cycle. Thermal efficiency depends on compression. ratio (r) and specific heat ratio (y):

$$\eta_{\text{thermal}} = 1 - \frac{1}{r^{\gamma-1}}$$

Even with lean-burn hydrogen combustion, thermal losses to coolant & exhaust cap real-world cycle efficiency at 35-40%.

Result: High Fuel Consumption (~1.18 kg H2 / 100 km)

PEMFC: ELECTROCHEMICAL ADVANTAGE

Not bound by Carnot limitations. Governed by Gibbs Free Energy (ΔG):

$$\eta_{FC} = \frac{\Delta G}{\Delta H} = 1 - \frac{T\Delta S}{\triangle H}$$

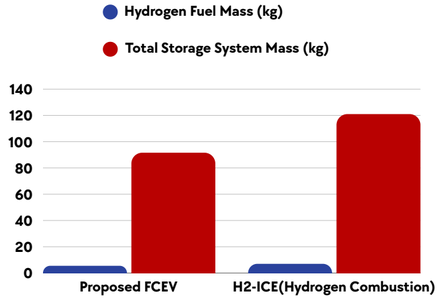
Direct conversion of chemical energy to electrical energy. Our system achieves validated 48.1% Tank-to-Wheel efficiency.

Result: Low Fuel Consumption (~0.92 kg H2 / 100 km)

22% reduction in onboard hydrogen mass for the same 597 km range

The D-Segment Packaging Crisis:

Hydrogen Storage Required for 600 km Range



EMISSIONS CHEMISTRY

H2-ICE: THERMAL NOx FORMATION (ZELDOVICH MECHANISM)

Hydrogen burns at high temperature in ambient air ($N_2 + O_2$).
Triggers thermal NOx formation:

$$O + N_2 \rightleftharpoons NO + N \qquad N + O_2 \rightleftharpoons NO + O$$

Requires heavy aftertreatment: SCR + Urea Dosing

PEMFC: PURE ELECTROCHEMICAL REACTION

Operates at low 60-80°C electrochemical boundary.

$$2H_2 + O_2 \rightarrow 2H_2O + \text{Electricity} + \text{Heat}$$

Result: Zero Tailpipe Emissions (Pure Water Vapor & Heat)

The Engineering Justification(Why not H2-ICE?)

The Suzuki Context

Recent prototypes (like the 2026 Suzuki Swift H2-ICE) prove hydrogen combustion is an excellent, low-CAPEX retrofit for lightweight hatchbacks and two-wheelers.

The D-Segment Thermodynamic Wall

Scaling H2-ICE to a heavy 1,950 kg GVW sedan creates a volumetric packaging crisis. H_2-ICE consumes significantly more kg H_2/100 km than a fuel cell.

The Packaging Verdict

At the same vehicle-dynamics basis, H_2-ICE requires ~7.0 kg H_2 for 597 km versus 5.5 kg for the FCEV. The PEMFC achieves the same range on just 5.5 kg of hydrogen due to its superior 48.1% TTW efficiency.

Emissions Reality

H_2-ICE: burns hydrogen in ambient air, producing NOx that requires complex SCR catalysts and urea dosing. PEMFC is a pure electrochemical reaction yielding only water.

Application Segmentation: THE SUZUKI PATH FORWARD

H2-ICE: The Urban Retrofit Solution

- 2-Wheelers (e.g., Burgman H2)
- Small Hatchbacks (e.g., Swift H2-ICE)
- Light Commercial Vehicles
- Low CAPEX, Fast to Market




Lightweight. Lower Range. Cost-Effective.

PEMFC: The D-Segment Solution

- D-Segment Sedans (1,950 kg GVW)
- Long-Range (500+ km)
- Premium Passenger Fleet Vehicles
- Efficiency, Space & Zero Emissions



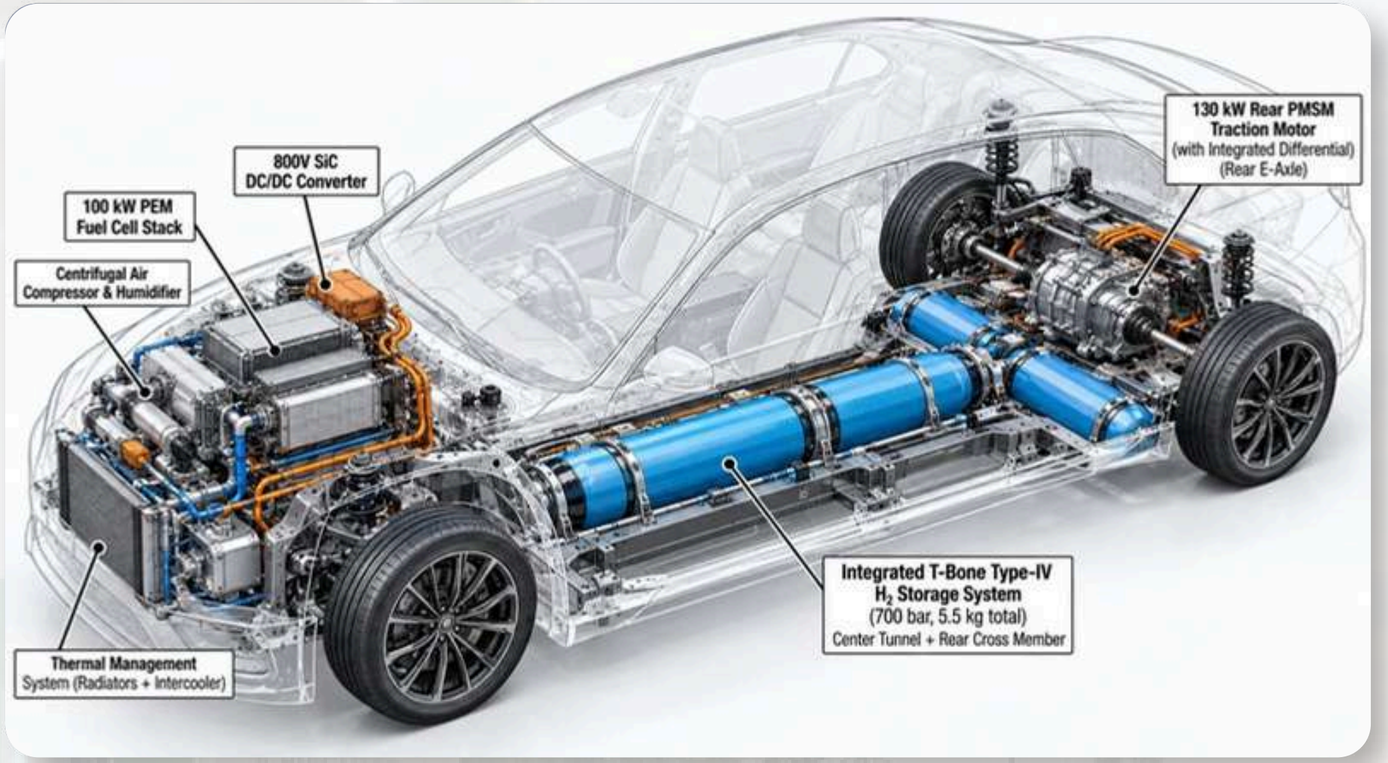
Heavyweight. Long Range. Space. Zero-NOx.

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Vehicle Architecture: Hybrid PEM FCEV

100 kW PEMFC + 700-bar Type IV H_2 + 1.5 kWh Buffer + 800V SiC Electric Drive

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Subsystem Architecture & Engineering Justification

Subsystem	Selected Architecture	Key Specifications	Engineering Justification
Dual-Loop Thermal System	Split HT/LT cooling with DI coolant	HT: **65–80°C** (Stack)LT: **<50°C** (Motor)	Independent cooling improves reliability while reducing radiator size and system mass.
Traction Motor & Drivetrain	**130 kW PMSM** + single-speed gearbox	0–14,000 rpm>95% motor efficiency	Meets **0–100 km/h in 8.7 s**, eliminates multi-speed transmission, saving **35–45 kg**.
800 V SiC DC/DC Converter	Bidirectional SiC converter	250–400 V → 650–800V **98% efficiency**	Halves current, cuts **I^2R losses by ~75%**, and stabilizes the DC bus during peak loads.

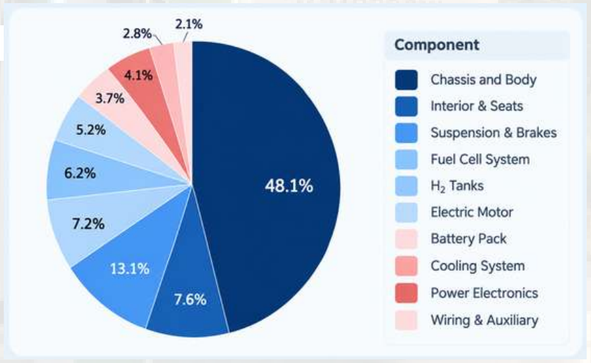
TARGET ENGINEERING SPECIFICATIONS

Subsystem	Selected Value	Engineering Justification
Fuel Cell Stack	**100 kW PEMFC**	~15.4 kW first-principles validation.
H_2 Storage	**5.5 kg @ 700 bar**	597 km range, checked by two independent methods that vary by 4%
Battery Buffer	**1.5 kWh (NMC/LTO)**	Bridges the 30 kW peak gap (100→130 kW) with 20C NMC/LTO capability.
Traction Motor	**130 kW PMSM**	0–100 km/h in 8.7 s with ~50% power margin.
Power Bus	**800 V SiC Architecture**	Halves current and cuts I^2R losses by ~75% vs. 400 V.
Vehicle Mass	**1454 kg Curb / 1950 kg GVW**	**496 kg payload budget; supports 5 occupants + ~120 kg luggage, subject to final design target.**

Membrane Technology Trade-Off: PEMFC vs. AEM

Parameter	Vehicle Fuel Cell (PEM)	H2 Production (AEM Electrolyzer)
Technology	**PEMFC**	**AEM**
Why Chosen	High power density & fast response	Low-cost, scalable hydrogen production
Why Not the Alternative	AEMFC: Lower power density, CO_2 sensitivity	PEM: High Ir/Pt & titanium cost
Operating Ion	H^+ (Proton)	OH^- (Hydroxide)
Catalyst	Low-loading Platinum	PGM-free (Ni/Fe/Co)
Performance	1.8–2.2 W/cm², instant load response	1.0–1.5 A/cm², steady MW-scale operation
Lifetime	5,000–8,000 hrs	< 2,000 hrs
Cost Impact	Optimized for compact 100 kW stack	~40–50% lower CAPEX than PEM
Best Suited For	Lightweight, high-performance vehicles	Cost-effective, large-scale H_2 production

Mass Breakdown & Weight Distribution Budget



BENCHMARK STUDY FRAMEWORK



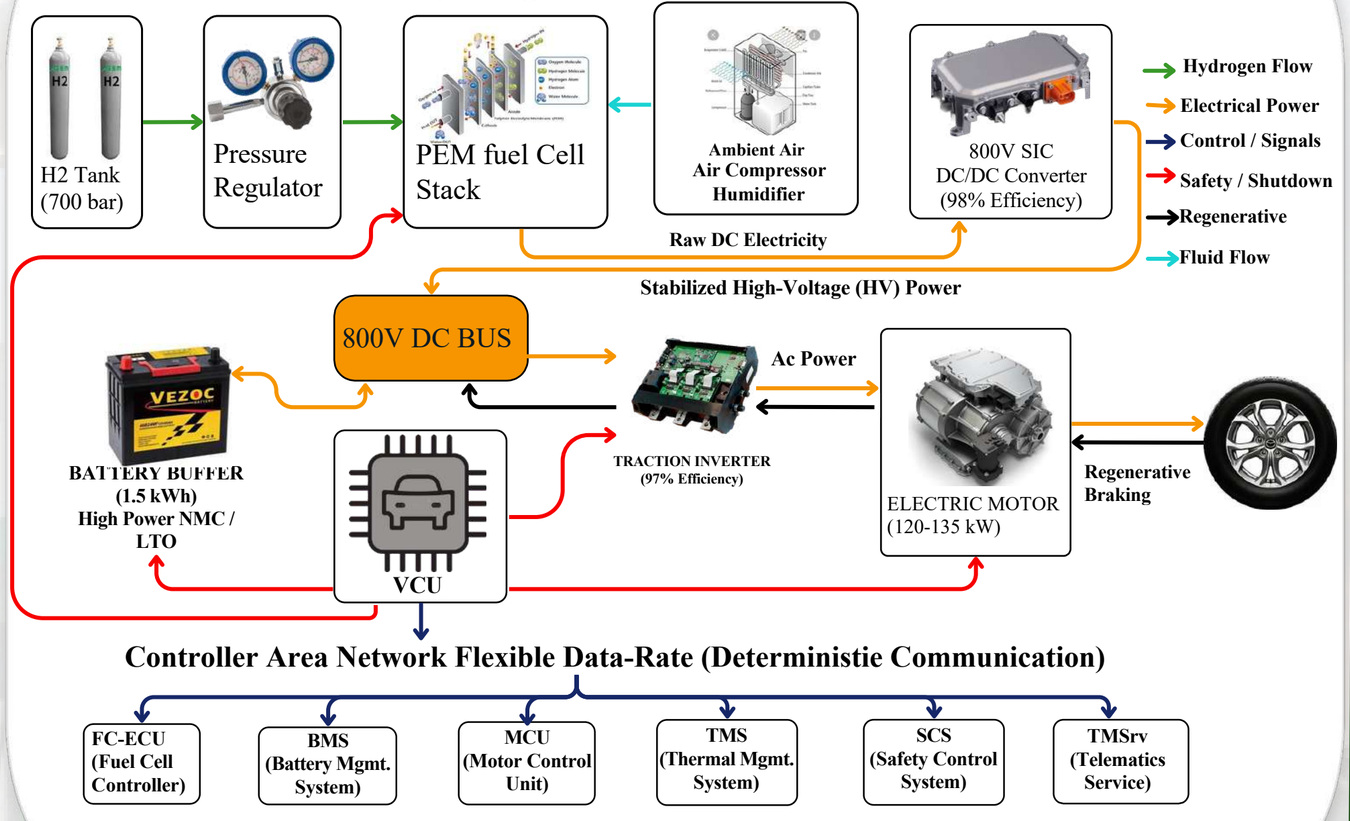
Benchmark study is to analyze globally proven hydrogen fuel cell passenger vehicles, identify best engineering practices, understand design trade-offs, and extract engineering lessons that can be adopted in the proposed vehicle concept.

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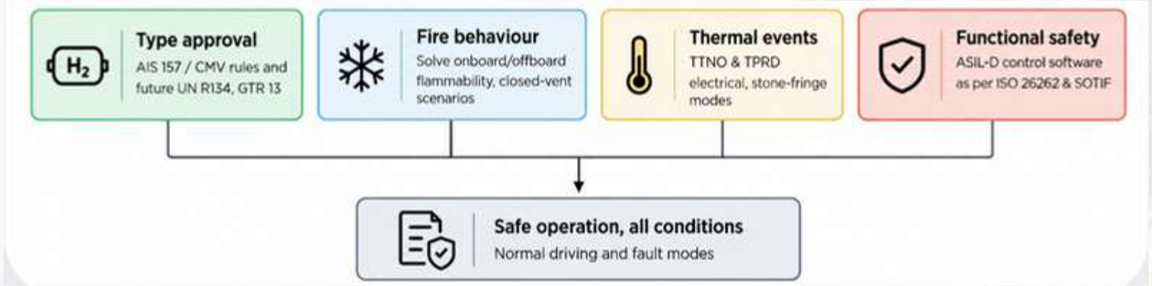
Powertrain Control & ASIL-D Safety Architecture

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800 V SiC Hydrogen FCEV Powertrain Architecture



Functional Safety Requirement Architecture



From Fault to Safe State

Layered Safety Response for a Hydrogen FCEV

1.DETECT

Identify a fault condition

- H_2 leak (pressure/concentration)
- Crash event (airbag / inertial sensor)
- HV isolation fault (insulation / short)
- Thermal event (over-temperature)

2.ISOLATE

Immediately remove energy sources

- Close tank valves.
- Open HV contactors
- Stop H_2 supply to fuel cell
- Isolate high-pressure lines

3.CONTAIN

Prevent escalation

- Prevent continued fuel delivery.
- Protect HV bus and components.
- Limit fault propagation.
- (electrical, thermal, mechanical)
- Activate ventilation (if required)

4.SAFE STATE

Bring the vehicle to a controlled state

- Controlled shutdown of powertrain
- Hazardous-energy isolation
- (H2, HV, thermal)
- Maintain system stability

The 1.5 kWh buffer is for regenerative braking and peak power, for genuinely critical faults skip straight to a fail-safe stop rather than trying to keep driving.

Our smart controls and safety architecture combine a deterministic CAN-FD network, hydrogen isolation set at real flammability limits, ASIL-D functional safety

The design principle is simple: detect early, isolate locally, and "fail safe-not keep driving at all costs."

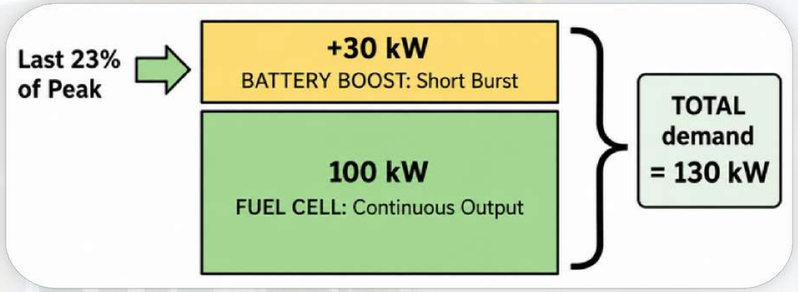
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Core Powertrain & Thermodynamics

100 kW PEMFC + 1.5 kWh buffer + 130 kW PMSM | 5.5 kg H_2 | 15.43 kW cruise demand at 100 km/h

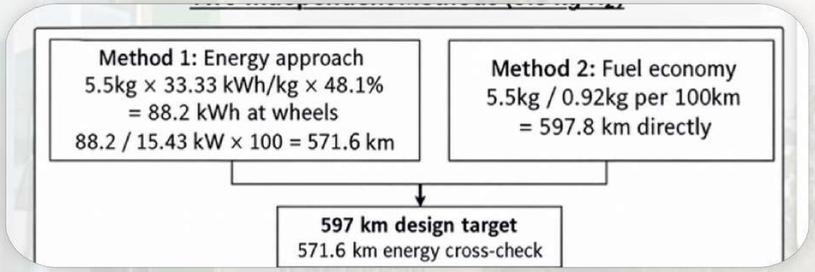
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How a 130KW peak motor demand is met?

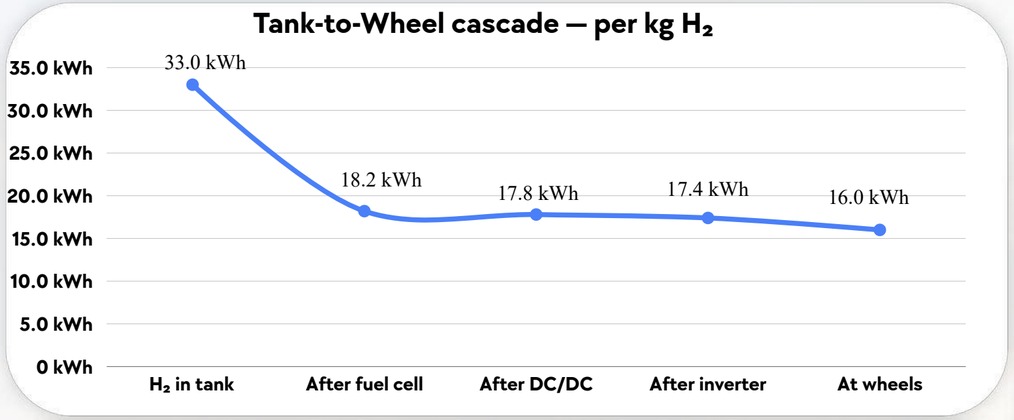


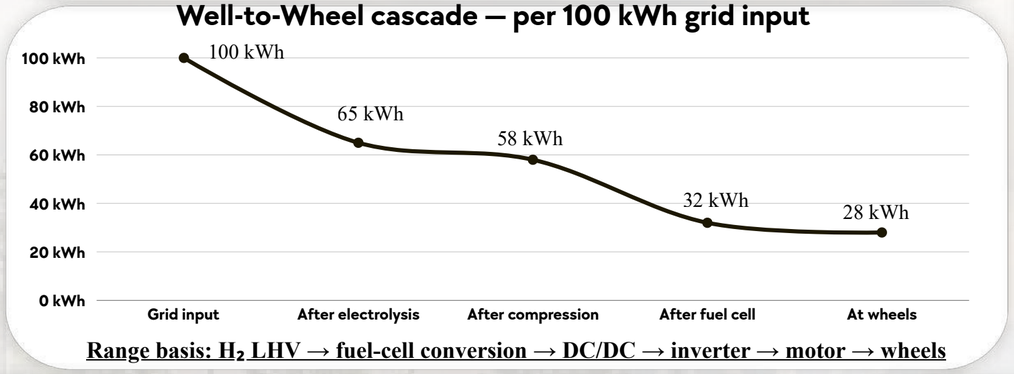
The 1.5 kWh buffer's job is defined by a real power gap (130 − 100 = 30 kW), with a realistic 50–60% usable window it is about 1.5–2 minutes of continuous coverage - far more than any single acceleration event needs.

Two Independent check methods(597km)



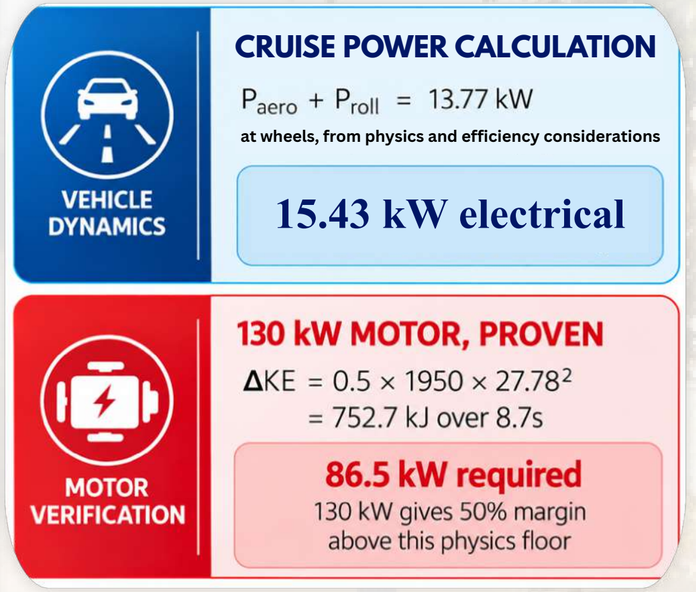
PEMFCEV EFFICIENCY





Range basis: H_2 LHV → fuel-cell conversion → DC/DC → inverter → motor → wheels

VERIFICATION OF VEHICLE DYNAMICS AND POWER SYSTEMS



Thermal Rejection Load

- **Fuel Cell Output**: 100 kW electrical
- **Efficiency**: 55%
- **Waste Heat to Reject**: 81.8 kW
- **Solution**: Dual-loop deionized water cooling system engineered to prevent stack derating at 45°C Indian summer ambients.

1. Environmental Derating & Real-World Headroom

- **Altitude Compensation:** Stack sized for operation up to 3,500 m MSL (e.g., Leh corridor); centrifugal air compressor actively increases pressure ratio to offset reduced ambient air density.
- **High-Ambient Thermal Headroom:** Dual-loop cooling system engineered with an 85°C coolant derating threshold, handling extreme 45°C Indian summer ambient temperatures without stack thermal throttling.

2. Certification Gap & Homologation

- **Test Cycle Calibration:** Steady-state cruise yields 597 km, while dynamic stop-and-go WLTC/MIDC cycles typically derate baseline range by 10–20%.
- **Homologation Standard:** Final vehicle certification requires chassis dynamometer validation under AIS-040/ CMVR (incorporating UN GTR 13 and ISO 26262 functional safety protocols).

These numbers are not pulled out of thin air; they are constrained by calculations and real-world operating conditions!

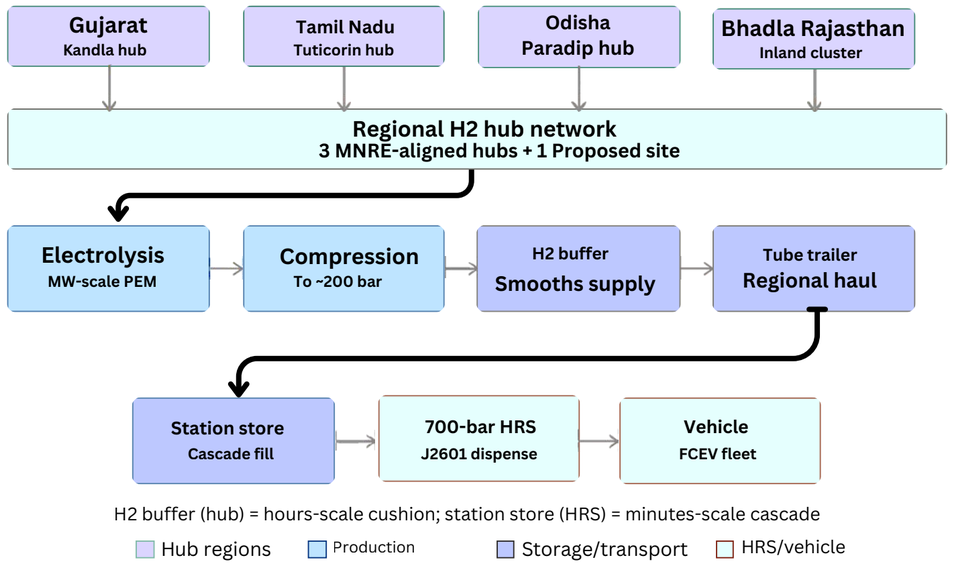
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Hydrogen Production & Supply Chain-Renewable Grid to 700-Bar Refuelling

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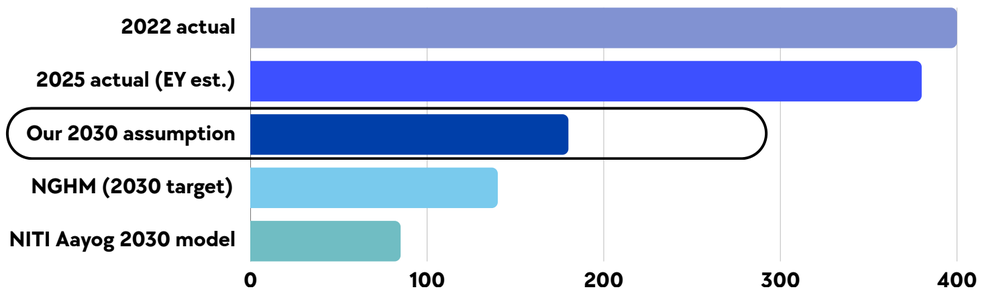
Regional H_2 production hubs aligned with India's Green Hydrogen Mission | 5.5 kg H_2 → 597 km range | ₹19,744 Cr mission-backed roadmap

From Renewable Power to the Road



India's hydrogen supply chain isn't one hub-it's three MNRE-aligned and one proposed regional hubs, each running the same electrolysis-to-refuelling pipeline, with two distinct storage layers that shouldn't be mixed.

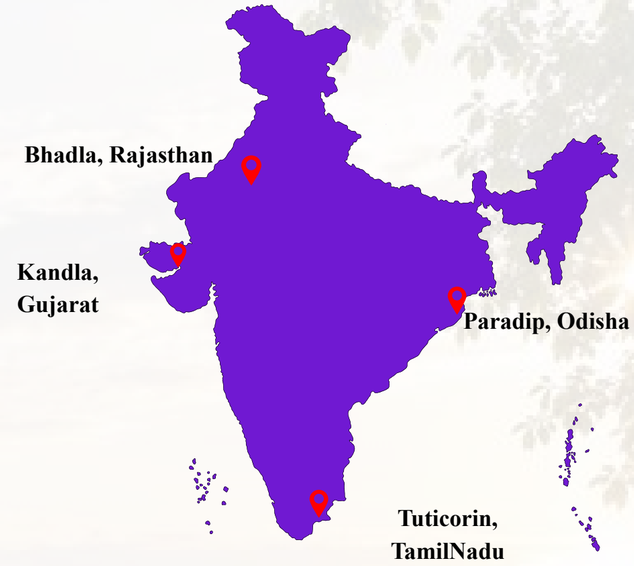
GREEN H2 COST: REPORTED vs MODELLED vs OUR ASSUMPTION



Our ₹180/kg assumption isn't invented - it sits between NGHM own modelled 2030 best-case (₹140/kg) and the NITI Aayog target (₹89/kg), while today's actual reported cost is more than double that.

WHAT THIS MEANS FOR OUR FCEV

1. **4 regional hubs** enable distributed H_2 production rather than dependence on a single source.
2. **Two-stage storage:** The hub buffer absorbs renewable variability; station cascade storage enables rapid vehicle filling.
3. This assumption directly gives a **₹1.66/km fuel cost**, used in our 2030 TCO calculation.



Hydrogen Cost Breakdown (2030 Target)

Multiple drivers reduce the delivered cost of hycrogen, making ₹180/kg a defensible, prudent design assumption.

1. Plant-Gate Production (~₹110- 125/kg)

- Backed by NITI Aayog's 2030 model ($1.3 - 1.5/kg)
- Uses dedicated ultra-low tariff solar/wind (<₹2.2/kWh)
- Supported by domestic electrolyzers under the ₹19,744 Cr NGHM

₹110-125/kg (at plant-gate)

2. Regional Haul & Compression (~₹35 - 40/kg)

- Bulk compression to 200 - 250 bar
- Tube-trailer transport within 150 -250 km radius corridors from regional hubs

₹35-40/kg

3. Station Dispensing & Margins (~₹20- 25/kg)

- Amortized 700-bar cascade storage
- T40 chiller precooling
- Dispenser CAPEX shared across passenger and commercial fleets

₹20-25/kg

Total Hydrogen Cost (Plant-Gate to Station)

~₹170-190/kg
(for current scenario)

Why Conservative?
0.92 kg/100 km×₹180/kg=₹165.6/100 km.
(0.92×180)/100=₹1.656/km

₹1.66/km

AEM & PEM

- **Electrolysis Strategy:** PEM Today, AEM Tomorrow
- **2030 Cost Driver:** Transitioning to Anion Exchange Membrane (AEM) electrolysis removes expensive precious metals from the stack, accelerating the drop toward ₹180/kg.
- **Immediate Baseline:** Current production relies on proven PEM electrolysis backed by the ₹19,744 Cr National Green Hydrogen Mission.

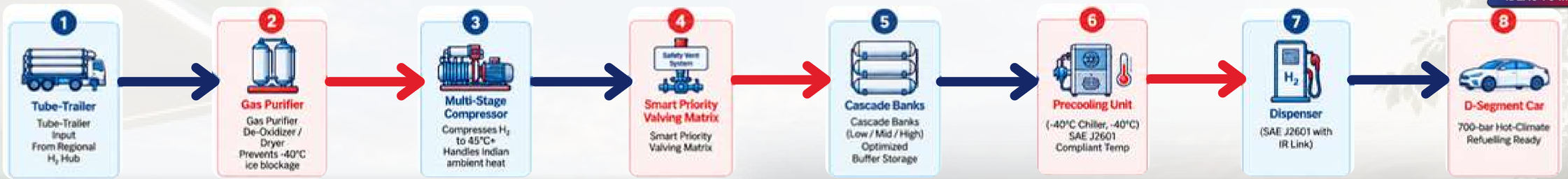
The strategy shows that hydrogen availability accompanied by a decentralized refueling infrastructure will result in FCEV becoming a scalable and realistic mobility pathway.

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Refuelling Infrastructure: 700-Bar HRS Architecture

Buffered before demand • SAE J2601-compliant dispensing • Designed for Indian summer conditions

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Hydrogen is stored before the vehicle arrives

- Compression ⟶ Cascade storage ⟶ Pre-cooling ⟶ 700-bar dispensing
- High-pressure H_2 is buffered in cascade banks, so filling does not depend on real-time compressor output.
- Low ⟶ medium ⟶ high bank sequencing maintains pressure and enables fast fills.

Less than 5 min is an engineered target

- 5.5 kg H_2 / 5 min = 1.1 kg/min (66 kg/hr)
- SAE J2601 reference: 3 - 5 min at T40 - 1.83 kg/min
- A 5-min target deliberately includes operating margin.

Indian ambient conditions are designed into the HRS.

- Hub states can reach 40-45°C in summer.
- Higher ambient temperature can derate filling performance.
- T40 pre-cooling + real-time MC-formula control dynamically manages fill rate and tank temperature.

Technology is already entering India

- Leh: operational green H_2 station-80 kg/day, 1.7 MW solar
- Kandla: HRS under tender with 18 kg/hr electrolyser
- National pilots: 9 HRS across 10 routes
- Indicative HRS capex: ₹10-20 Cr/station





Fig. Hydrogen Refueling Stations

Leak Warning Limits: Station-to-vehicle dispensing uses strict volume monitoring: a warning sounds at 2% hydrogen volume in the air, and an emergency cutoff triggers instantly at 4% volume.

High Auto-Ignition Safety: Hydrogen needs a very high surface temperature of 585°C to catch fire on its own, which is far safer than petrol vapors that ignite at just 246°C.

- **An IR link enables real-time tank temperature monitoring, allowing the dispenser to dynamically adjust H_2 flow at 700 bar for safe, optimized filling in hot climates; communication failure triggers a slower fallback fill rate.**

- **Fuelling mechanics and tank durability meet international standards SAE J2579, UN GTR 13, and ISO 19881 for crash and pressure safety**

Compress, cascade storage, pre-cool, dispense-our HRS is engineered for fast refuelling in India's climate, using infrastructure India is already deploying.

The technology stack is not hypothetical: India is already commissioning hydrogen mobility projects. Our design scales that ecosystem to a 700-bar passenger-FCEV architecture.

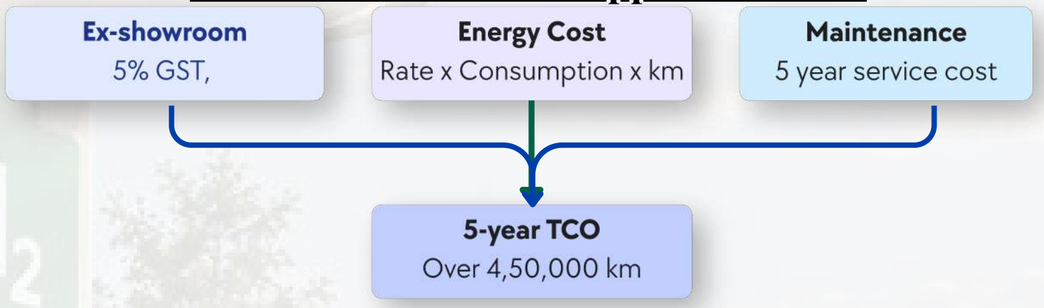
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Economics-5-Year TCO, Verified and Stress-Tested

5-year FCEV TCO: ₹69.43L → ₹52.63L by 2030 - a ~24% reduction, evaluated on a real 300-350 km/day taxi duty cycle

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Total Cost Of Ownershipp Breakdown



Parameter	Long range BEV (BYD Seal)	BEV Baseline	FCEV (current)	FCEV (2030)
Ex-showroom	₹45.55L(610km)	₹25L	₹48.0L	₹40.5L
Energy cost	₹10.9L	₹10.9L	₹17.04L	₹8.07L
5-yr Maintenance	₹3.90L	₹3.90L	₹4.39L	₹4.06L
Replacements	₹15.0L	₹5.0L	-	-
5-yr TCO	**₹75.35L**	**₹44.80L**	**₹69.43L**	**₹52.63L**

8,000 hours is a durability target—not a failure point. DOE defines durability using the time to 10% voltage degradation; this metric may not represent end-of-life and does not address catastrophic failure modes

2030 Cost-Reduction Levers

1 Localized Dome Reinforcement via FPP



Apply carbon fiber patches only to the high-stress dome end-caps using **Automated Fiber Patch Placement (FPP)** instead of full-length wrapping.

Cost lever: Tank material & manufacturing cost

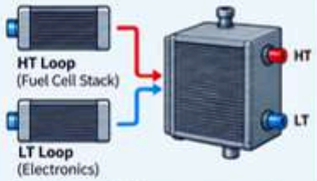
2 Lower the Tank Cutoff Pressure Threshold



Program the **powertrain software** to safely draw hydrogen down to **5 – 10 bar**, reducing unextractable cushion gas and allowing **~5% smaller tanks** for the same 600 km range.

Cost lever: Tank system cost

3 Co-Brazed Dual-Circuit Microchannel Radiator



Combine the high-temperature stack loop and low-temperature electronics loop into a single co-brazed radiator with split internal fluid paths.

Cost lever: Thermal system cost

4 Leverage SIGHT Scheme Incentives

Government of India
SIGHT
STRATEGIC INTERVENTIONS FOR GREEN HYDROGEN TRANSITION

Set up fuel cell integration assembly within India's designated green hubs to tap into the ₹4,440-crore SIGHT electrolyzer and fuel cell localization pool.

Enabler: Lower overall vehicle and system cost

Fuel-Cell Lifetime-Extension Techniques

Technique	Implementation	Durability Benefit
Eliminate OCV Dwell	Avoid high-voltage, near-zero-current idling.	Reduces **Pt dissolution & catalyst degradation**.
Minimise Start–Stop Cycles	Use **controlled purge + minimum unnecessary shutdowns**.	Prevents **reverse-current spikes & carbon corrosion**.
Lower Current Density	Increase active area for the same stack power.	Reduces **membrane/catalyst stress & voltage decay**.
Battery Load Smoothing	1.5 kWh buffer absorbs rapid power transients.	Reduces **OCV exposure & load-cycle degradation**.
Thermal & Water Management	Maintain optimum temperature/hydration; prevent flooding.	Limits **flooding-induced carbon/support degradation**.
Degradation Monitoring + Derating	Track stack voltage/impedance and progressively shift load to battery.	**Extends usable service life** as the stack ages.

Why not Stack Refurbishment Line

Argument	Simple Technical Reason	TCO Impact
Performance Recovery	Some voltage loss is temporary and can be recovered through **purging and water management**.	**₹0 additional hardware cost**
Battery Compensation	The **1.5 kWh battery** supports the motor as the stack ages, maintaining required vehicle power.	**No planned TCO impact**
Component-Level Service	If degradation is localized, **affected cells/components can be serviced** instead of replacing the complete stack.	**Potentially ₹0.3–0.5L** vs. ₹5.5L full replacement
OEM Warranty Support	Commercial FCEVs can be covered by **multi-year/high-mileage fuel-cell warranties**.	**Potentially ₹0 operator cost** within warranty

Our FCEV reaches cost parity with the premium long-range BEV under real intercity-taxi conditions, needs no stack replacement within 5 years, and stays within a defensible range even under worst-case hydrogen pricing.

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2030 Implementation Roadmap & Conclusion

Two verified phases. Three converging factors. One clear role for FCEVs.

Igniters Innovation Challenge 2026
IDEAS TO IMPACT

2026 Pilot launch → **2028** Scale-up begins → **2030** Target state

Phase 1: Corridor HRS

₹496 Cr scheme total outlay up to FY 2025-26, with ₹208 Cr of central support specifically sanctioned across 5 consortiums for 37 buses and trucks.
9 HRS (Hydrogen Refueling Stations) approved.

Phase 2: nationwide scale

~100 HRS by 2030
D-segment MULTI UTILITY FCEV entry (fleet & intercity taxi)

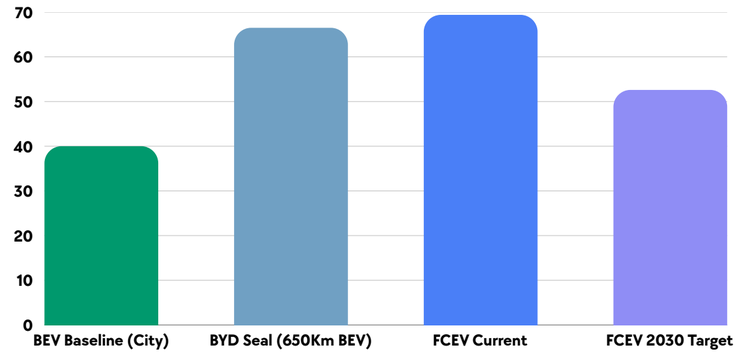
2030 target state: three factors converge

- **Customer uptime** — Under 5-min refuel
- **Cost Reduction for Hydrogen Vehicle**
- **National policy & regulation** — NGHM, H2 at ₹140/kg; Vehicle and station type-approval already active (AIS, PESO)

2030 compatibility vision

FCEVs complement BEVs, not replace them

Economics strengthen with scale



5-year FCEV TCO:
₹69.43L - ₹52.63L by 2030

Remains cost-competitive with a comparable long-range BEV

2030 Implementation Roadmap & Conclusion

Infrastructure is moving from pilot to scale

- **2026 pilot**: 10 routes • 37 vehicles • 9 HRS approved
- **2030 direction**: ~1,000 hydrogen High utility intercity vehicles; ~100 HRS as planning scale.
- Our vehicle is designed around this emerging infrastructure, not a hypothetical network.

Range comfortably supports corridor operation

- 597 km range vs 200 km planned HRS spacing
- One full tank clears two consecutive station gaps (400 km) with ~197 km range remaining
- **Result:** planned corridor spacing does not constrain a full-tank trip.

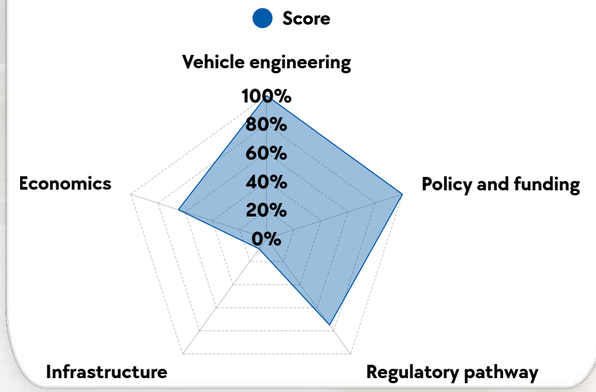
Economics strengthens with scale.

- 5-year FCEV TCO: ₹69.43L → ₹52.63L by 2030
- Remains cost-competitive with a comparable long-range BEV
- FCEV cost is reduced by amortizing infrastructure and lower cost H2 production

The 2030 role is complementary.

- BEVs: strong fit for shorter-range, charging-accessible use cases
- **FCEVs:** strong fit where long range + fast refuelling + high utilisation matter
- **Conclusion:** FCEVs complement BEVs rather than compete with them everywhere.

Technology Readiness Level (TRL) & Commercialization Viability Index



"Government infrastructure is being built for heavy-duty hydrogen fleets. Our passenger FCEV shares the hydrogen supply chain; each station needs a 700 bar dispenser add-on."

SUZUKI
Suzuki R&D Center India Pvt. Ltd.

Igniters Innovation Challenge 2026

IDEAS TO IMPACT



Thank You

Driving India towards a cleaner, smarter and hydrogen-powered future.

TECHNICAL ANNEXURE

Detailed calculations, assumptions, validation & engineering analysis

https://drive.google.com/file/d/1LEwTlJl-RnK4jUGIhA6KWgn4fSjq8kXS/view?usp=sharing